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High-resolution characterization of ceramic-metal interface of TiN coating on ferritic-steels for nuclear application

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Abstract

Advanced fuel cladding is critical for fast reactors, offering sufficient thermal conductivity, mechanical and dimensional stability and radiation tolerance of the cladding base material. Additionally, it must provide corrosion resistance and high temperature coolant compatibility on the cladding outer surface, as well as chemical stability on the cladding inner wall against fuel cladding chemical interaction (FCCI). TiN ceramic coating has been considered an effective diffusion barrier for inner and outer cladding-walls for enhanced performance. The TiN-metal interface microstructure and chemistry play a critical role in coating bond strength and integrity under harsh conditions. High-resolution transmission electron microscopy characterization of ceramic-metal interface at atomic resolution in unirradiated, irradiated and thermal

Outline

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cycled conditions were performed. The interface remained intact after irradiation up to 200 dpa or thermal cycling five times up to 550 °C. This work discusses the potential impact of these results on coating performance and design for advanced claddings.

Keywords

Cladding; Fuel cladding chemical interaction; Ceramic coating; Interface; Irradiation

Acknowledgement

Data availability

References

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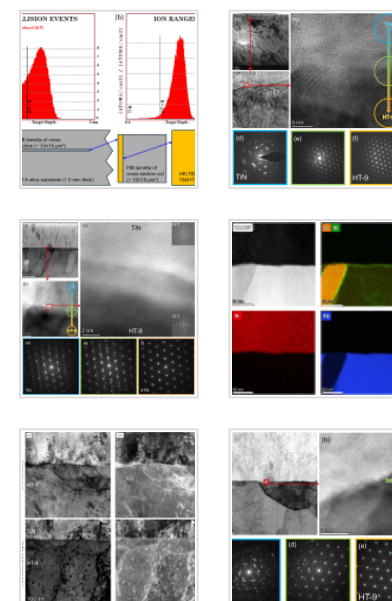
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1. Introduction

Fuel cladding is a critical component of nuclear fuel systems for the safe operation of nuclear reactors. Fuel cladding chemical interaction (FCCI) can severely affect the cladding integrity and lead to the premature failure of fuel elements. There are several comprehensive reviews on FCCI summarizing the extensive irradiation experiments and fundamental aspects of materials degradation in both metallic fuels and claddings [[1], [2], [3]]. There is an increasing need for R&D on the advanced claddings to mitigate the FCCI for high burnup applications particularly for innovative nuclear power systems. Fast spectrum nuclear reactors are crucial for the sustainable development of nuclear energy systems. They enable the effective utilization of uranium resources and the rational management of spent nuclear fuels, including advanced fuel cycles. Reprocessing spent nuclear fuels using fast reactors offers several advantages including significant waste volume reduction, reduced long-term radiotoxicity of the residual waste, enhanced energy efficiency and utilization, enhanced non-proliferation, and support for advanced reactor fuel R&D with closed loop of fuel life cycle. In 2023 an innovative cladding materials research and development project for advanced fast reactors was initiated under the Innovative Nuclear Materials R&D Program launched by the United States Department of Energy (DOE) – Office of Nuclear Energy. It calls for the mechanistic

Figures (8)



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understanding of materials challenges experienced with the traditionally more engineering focused approaches to promote the R&D for the advanced claddings. Several recent publications highlighted the work under this program [[4], [5], [6]].

Cladding is a crucial part of the fuel system, used in extreme environments. It is challenging to develop a cladding base material that satisfies all the requirements for high burnup fuels. These requirements include excellent neutron transparency, good thermal conductivity and radiation tolerance, low activation, high dimensional stability, adequate mechanical strength for in-pile performance at high temperature and out-pile storage for decades. Additionally, the material must exhibit good compatibility with fuel constituents and fission products on FCCI mitigation at inner surface and corrosion resistance with coolant at cladding outer surface. It was hypothesized that the advanced cladding consisting of a 3-layer structure may be more practical and achievable with its base material satisfying the major requirements aforementioned and the modified inner and outer surface layers to mitigate the FCCI and corrosion, respectively. Nitride-based ceramic thin coating, particularly the titanium nitride (TiN), has been very effectively demonstrated as a diffusion barrier to mitigate the diffusion reaction between iron (Fe) and cerium (Ce) under irradiation to high dose (100 dpa) at 500 °C, representative to FCCI [7]. Improved radiation tolerance of TiN_{0.7} rather than TiN due to enhanced self-healing effect was reported by Xue et al [8]. TiN coating is one of the most common physical vapor depositions (PVD) widely used for surface protection and performance enhancement for cutting and drilling tools due to its high thermal and chemical stability, good thermal and electrical conductivity, high hardness (~ 2500 H_v), high melting point (~ 2950 °C), and excellent wear and corrosion resistance [9].

Among various coating techniques, particularly in the field of PVD, pulsed laser deposition (PLD) is distinguished by its superior coating quality. This technique benefits from an energetic plasma flux generated by laser ablation, allowing the ceramic coatings to be applied at room temperature while achieving theoretical density and reproducing the stoichiometry of the target material for the coating. An excimer laser operating in the ultraviolet (UV) range in pulsed mode is typically chosen for most PLD systems. The

ablated material forms a plasma plume and then condenses onto the substrate to form a thin film. All depositions are conducted under high vacuum (10^{-6} torr) to prevent oxidation and degradation. PLD has great potential for engineering scale applications [10]. Both inner and outer tube coatings have been demonstrated by PLD with rotating tube or target designs [11]. The essence of a robust ceramic coating on metallic cladding for FCCI mitigation comes from the ceramic-metal interface bonding strength and its integrity under high-dose irradiation and large thermal stress from the mismatch of thermal expansion coefficient between the ceramic coating and metal substrate. The mechanistic understanding of the interface adhesion between dissimilar materials was explored by Evens et al [12]. A more comprehensive review on the ceramic-metal interface structure and its relationship to atomic and meso-scale properties with both experimental and modeling investigation can be found in the reference [13].

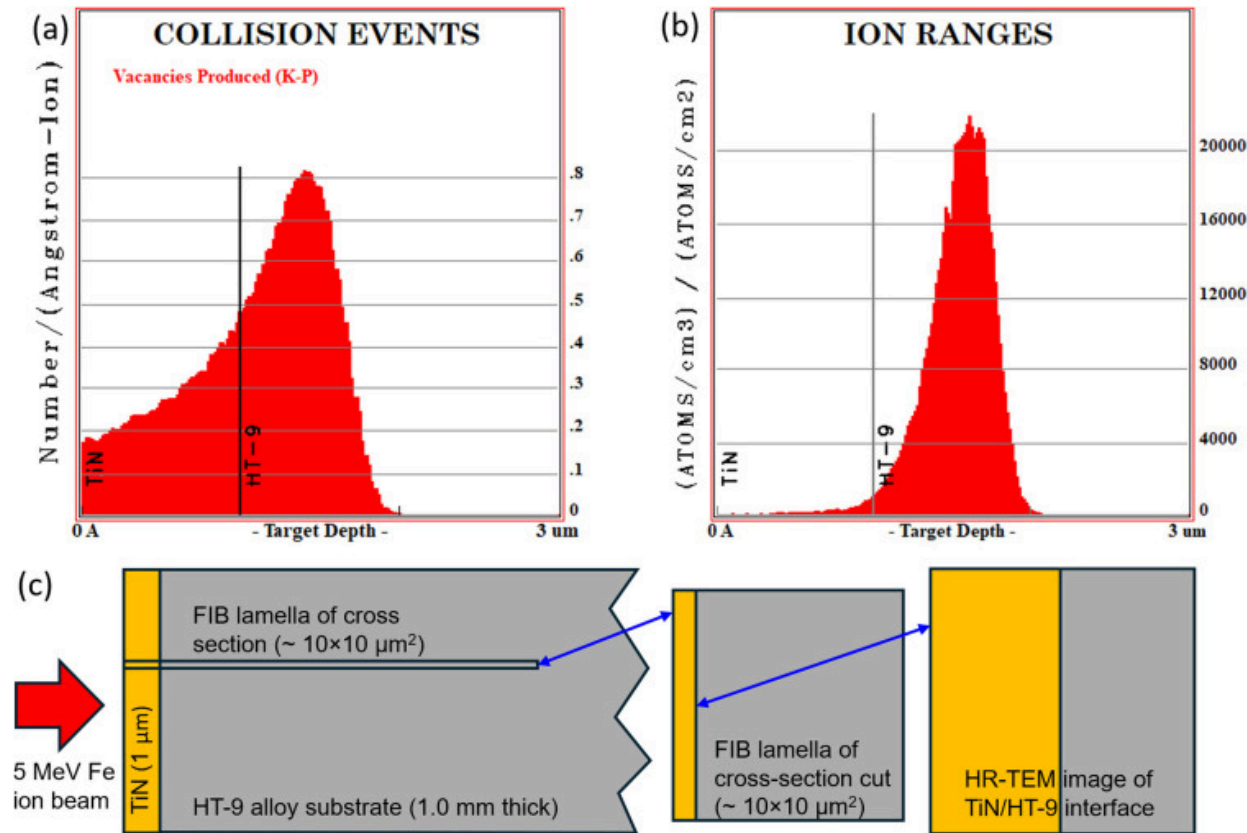
The atomic structure and chemistry at the ceramic and metal interface, including impurity segregation, can significantly influence bond strength and interface integrity. One fundamental question concerning ceramic and metallic cladding interfaces is how these interfaces accommodate a change in the atomic bonding types, from metallic bonds in metals to ionic or covalent bonds in ceramics. Another concern is how interface bonding could be affected by irradiation or thermal cycling. Comprehensive studies on ceramic-metal interfaces can be found in the literature [12,13]. The density function theory (DFT) calculations in our recent publication revealed the strong effect of stoichiometry of the TiN coating on the diffusion barrier strength [7]. Despite the previous studies demonstrating the strong bond strength of TiN coating on metal substrates using pulsed laser deposition (PLD) [11], atomic-level details at the ceramic-metal interface and the stability of the interface structure under irradiation and thermal cycling are still missing. It is generally understood that the ceramic-metal interface atomic structure and chemistry play an important role in the coating's bond strength and overall performance. In this work, we propose to conduct a high-resolution microscopy analysis at the ceramic-metal interfaces and explore the effect of the interface atomic structure and chemistry on the cladding's mesoscale performance. To our knowledge this

is the first study of cladding ceramic coatings to investigate the atomic structure at the ceramic-metal interface using high-resolution transmission electron microscopy (HR-TEM) analysis. The potential impacts of these results are discussed. Despite the fact that this is an initial effort, the findings and results lay a solid foundation towards a mechanistic understanding of the interface bonding at atomic scale and how that affects the mesoscale performance of ceramic coating for nuclear energy applications.

2. Experiment

There were four sample conditions investigated in this HR-TEM characterization work including (1) one sample prepared from the unirradiated region of an HT-9 alloy coupon coated with 1- μm thick TiN by PLD at a substrate temperature of 500 °C and (2) two samples prepared from the region irradiated with 5 MeV Fe ions at 500 °C to 50 and 200 dpa at the TiN-metal interface at the University of Michigan Ion Beam Laboratory, and (3) one 14-YWT oxide dispersion strengthened (ODS) alloy bar coated with 1- μm TiN by PLD at room temperature that underwent a thermal cycling test in vacuum (120–550 °C for 5 cycles with a temperature ramp rate of approximately 9.4 °C/min, 10^{-5} – 10^{-6} torr depending on temperature). The nominal compositions in wt. % for HT-9 and 14-YWT ODS alloy were Fe-12Cr-1.0Mo-0.5W-0.5Ni-0.25V-0.2C and Fe-13.6Cr-3.12W-0.4Ti-0.04Ni-0.02Mn-0.037Cu-0.008C-0.35Y₂O₃, respectively. Cross sectional TEM lamellas were prepared using focused ion beam (FIB) at the Center for Advanced Energy Research (CAES) at Idaho National Laboratory (INL). A Thermo Fisher Spectra 300 S/TEM operated at 300 kV was used for TEM characterization. All of the HR-TEM images were obtained without the objective aperture. The diffraction data at the ceramic-metal interface region was collected with a 10-nm selected area aperture. Limited composition analysis in the interface region was collected using SuperX in STEM mode. The TiN coatings were applied using a PLD system operated using a KrF Excimer laser with the wavelength of 248 nm (Lambda Physik Compex Pro 205) to ablate materials from a solid TiN target, under high vacuum of 1×10^{-7} Torr or better with the energy density in the range of 3 - 4 J/cm².

Heavy ion irradiation with 5 MeV Fe ions was conducted at the Michigan Ion Beam Laboratory (MIBL) to achieve the high dose at the interface region. Fig. 1 shows the radiation damage profile and injected Fe ion distribution as a function of ion penetration depth in the 1- μm TiN coated HT-9 coupon sample calculated using SRIM code with quick calculation option [14,15]. The density of 5.22 and 7.65 g/cm³ was used for TiN and HT-9 alloy, respectively. Atomic displacement energy of 25 and 28 eV was used for Ti and N, respectively, and 40 eV was used for Fe, Cr and Mo. A schematic of irradiation, FIB lift-out and HR-TEM analysis of the cross-sectional sample at the TiN-Metal interface is shown in Fig. 1(c). The ion irradiation was conducted at a substrate temperature of 500 °C with a rastering beam over an area of 10 × 16 mm². The estimated dose rate is approximately 5 × 10⁻⁴ dpa/s. At 200 dpa, the total ion fluence is 3.54 × 10¹⁷ ions/cm². The ion range plot revealed that the effect of injected interstitials on the irradiated microstructure at the TiN-Metal interface was expected to be minor with an estimated injected Fe ion concentration of approximately 0.41 % in HT-9 at the interface. High concentration of injected interstitials could promote loop formation and suppress cavity formation. The detailed discussion on the interstitial effect of the injected ions on the development of irradiated defect microstructure can be found in literature [16].



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Fig. 1. SRIM calculations of (a) radiation damage profile and (b) ion distribution along depth direction for an HT-9 coupon sample with 1- μm thick TiN coating by pulsed laser deposition. The schematic (c) illustrates ion irradiation, FIB lift-out, and HR-TEM analysis of the ceramic-metal interface.

3. Results and discussions

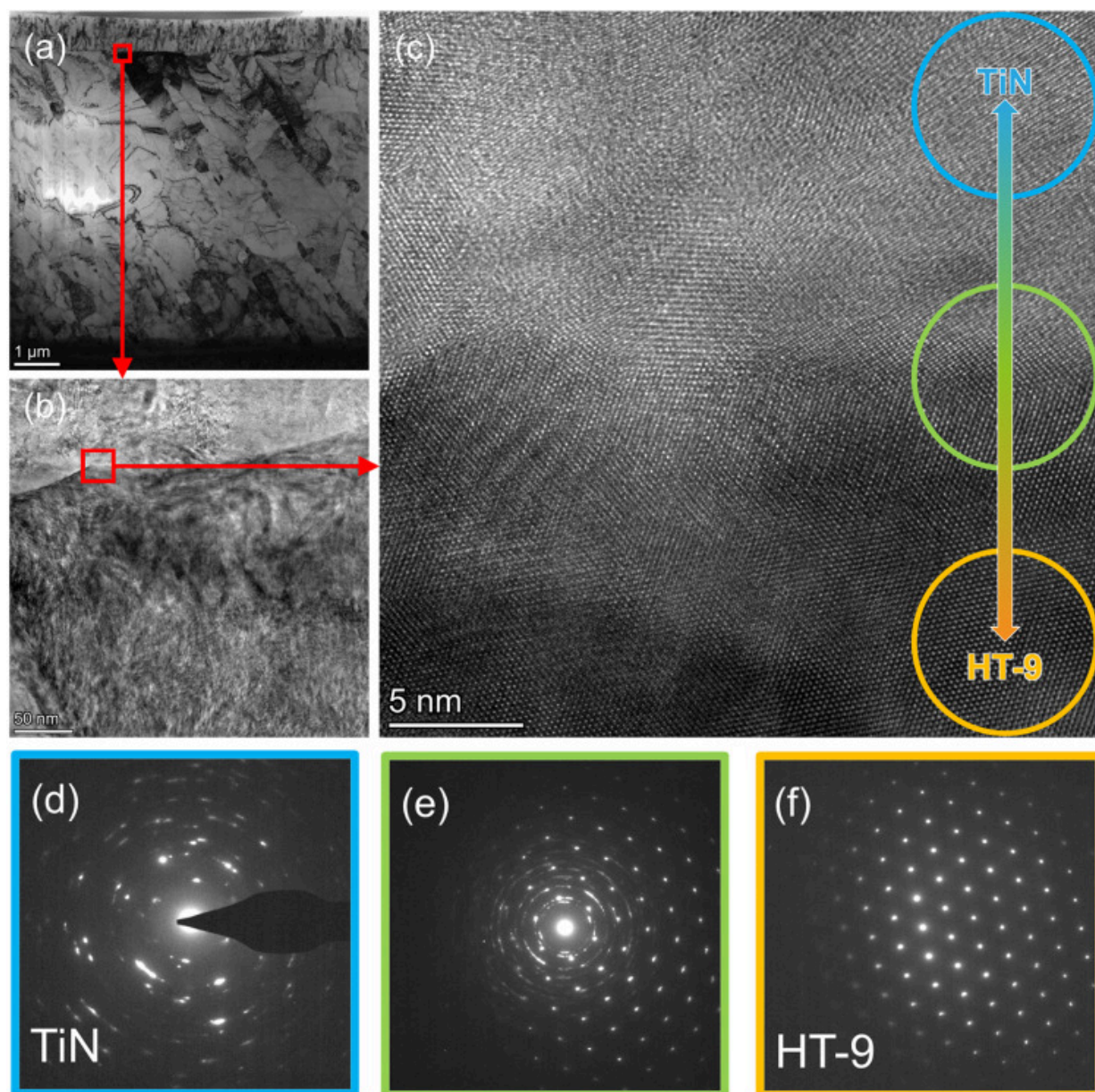
We performed focused ion beam (FIB) cut for cross sectional TEM sample preparation at the TiN-metal interfaces. This is due to the challenge in preparing the conventional cross-

sectional TEM samples via the cutting, grinding and polishing steps. The lack of atomic resolution TEM characterization on the interface microstructure of ceramic coatings in the open literature was likely due to the extreme challenges in the preparation of cross-sectional TEM sample for thin ceramic coating on metal substrate when FIB was not readily available. Note that FIB processing needs to be handled carefully to avoid potentially artifacts of (1) defect formation due to Ga ion damage and (2) defect removal due to annealing effect from local temperature rise from high ion current. With proper control of FIB parameters, both artifacts can be minimized and eliminated.

3.1. Unirradiated TiN-metal interface

The boxed area in Fig. 2 shows where the HR-TEM image was acquired at the ceramic-metal interface from a cross-sectional TEM sample. This TEM lamella was lifted-out using a FIB from the unirradiated region of a legacy sample (i.e., 1- μm TiN coated HT-9 alloy coupon irradiated at 500 °C to 50 dpa with 5 MeV Fe ions). The bright field TEM image of the cross-sectional view in Fig. 2(a-c) revealed a sharp ceramic-metal interface at low and intermediate magnifications. However, the high-resolution image showed a slightly blurry ceramic-metal interface, revealing the interface microstructure at atomic resolution. Note that for the TiN coating by pulsed laser deposition (PLD) it was challenging to find a ceramic-metal interface region with both sides oriented at major zones that also retained a perfect flat interface. This was because the HT-9 surface finish prior to the TiN coating was done at approximately 0.1 μm and the deposited TiN coating layer had nanosized columnar grains. Although the TEM foil thickness for HR-TEM was only few tens of nm, the interface shown in Fig. 2(c) indicates that it was not perfectly even and not under the exact edge-on condition, as evidenced by the presence of Morie fringes in the mid-left region due to crystal overlapping along the electron beam direction. The purpose of HR-TEM was to show the atomic resolution imaging of the interface by tilting the sample to the low index major zone for both metal and ceramic at the interface. Note that not all the interface regions can be selected for HR-TEM imaging unless both sides can be tilted to low index major zone at the same time. From the interface overview in low magnification TEM image, it is speculated that the atomic

configuration revealed from the HR-TEM image of the selected region should be maintained over extensive interface regions.



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Fig. 2. TEM bright field image of cross-sectional sample: (a) TiN coating on HT-9 substrate, (b) an enlarged view of the TiN/HT-9 interface region from the boxed area in a, (c) HR-TEM image showing atomic structure of the ceramic/metal interface from the boxed region in b with the atomic structure showing HT-9 oriented at $[111]$ zone direction. The corresponding selected area diffractions from the interface region of TiN, interface and HT-9 are shown in (d-f), respectively, where the elongated spots indicate potentially textured TiN film, internal stress and atomic plane distortion.

Three colored rings indicate the approximate locations where the selected area diffraction patterns were collected. Elongated diffraction spots were noticed in both the TiN and interface regions but not in HT-9, as shown in Fig. 2(d-f). These elongated diffraction spots indicate the local atomic plane distortions due to high internal stress. This could be related to strong atomic bonding at the interface between the two materials. There were no microcracks and separation found at the interface. Comparing the diffraction in Fig. 2(d) for TiN coating and Fig. 2(f) for HT-9 substrate, it clearly shows the TiN grains are smaller than those of HT-9 as TiN grains are textured polycrystalline and HT-9 grains are nearly single crystal in the corresponding imaging areas. It is clear that the TiN and HT-9 interface formed atomic bonding regardless of the crystal orientation between the two materials although the exact local interface bond strength may vary depending on the crystal orientation between the two materials.

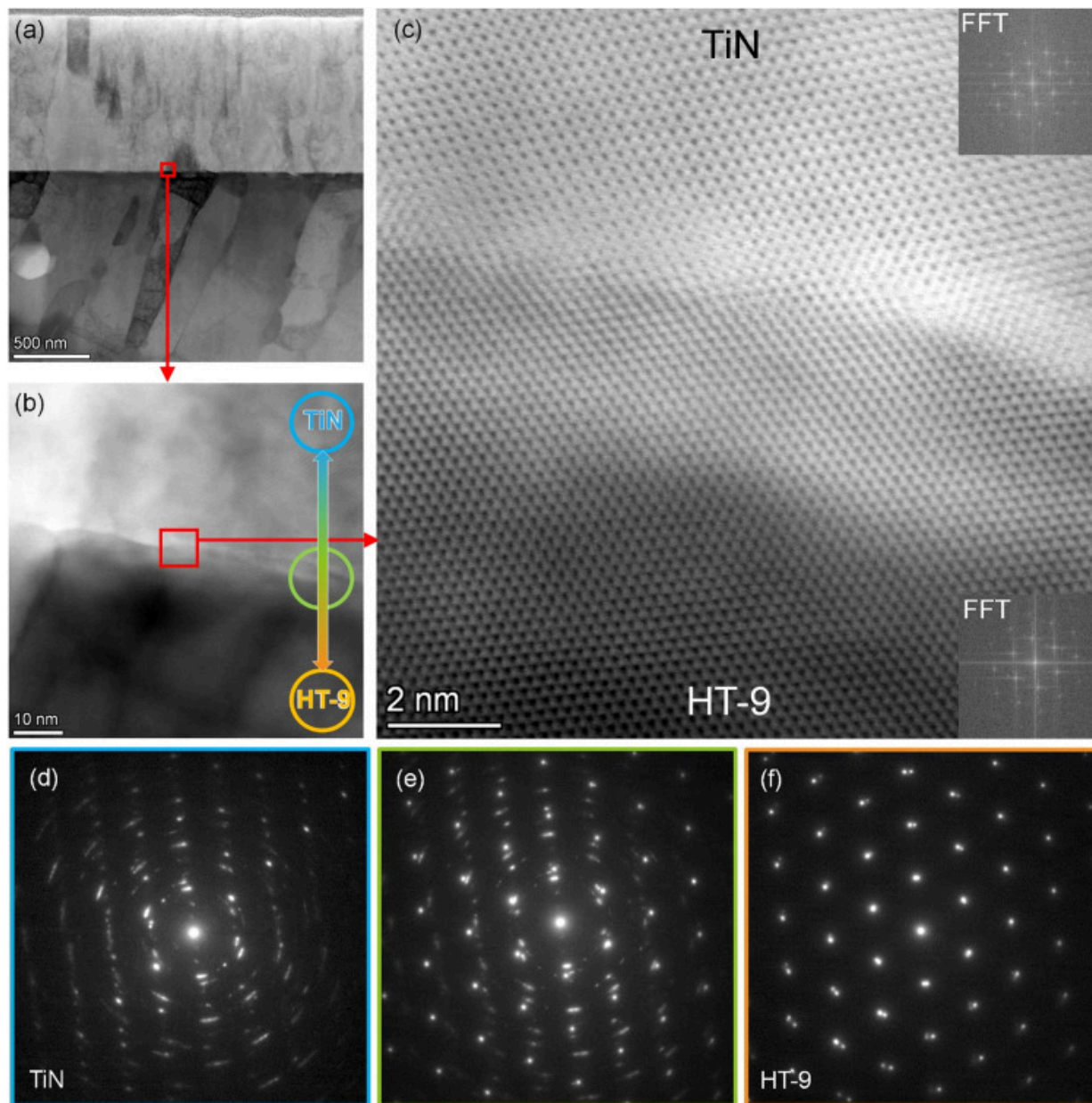
3.2. Atomic structure of TiN-metal interface irradiated to 50 dpa

Previous reports indicated that the TiN-Metal interfaces remained intact after high-dose irradiation, as observed through TEM characterization. However, the HR-TEM analysis at atomic resolution of the interface was lacking. It was speculated that irradiation could alter the interface atomic structure and chemistry by defect accumulation and elemental redistribution that could potentially weaken the interface bond strength. The previous section provided a reference for the unirradiated interface atomic structure from high-resolution (HR)-TEM analysis. This section provides the atomic-resolution interface microstructure and microchemistry after high-dose irradiation with 5 MeV Fe ions at 500

°C. Through the comparison, the irradiation effects on the detailed interface microstructure may be elucidated.

Fig. 3(a) and (b) reveal scanning transmission electron microscopy (STEM) bright field images of the cross-sectional view of the TiN/HT-9 interface at intermediate and high magnifications, respectively. The HR-STEM bright field image with atomic resolution from the boxed region in (b) is shown in Fig. 3(c) with HT-9 oriented near zone $[111]$ and TiN oriented near zone $[011]$ as shown by the Fast Fourier Transformation (FFT) in the two insets. Note that it is rare to find both the metal and TiN oriented along the low-index major zone at the interface for a FIB lamella. Surface oxidation of HT-9 alloy at the interface before PLD does not appear to be present in this particular region. The captured interface shows a smooth transition of atomic planes from $(01\bar{1})$ plane in HT-9 to $(\bar{1}1\bar{1})$ plane in TiN where the corresponding plane spacing for HT-9 and TiN was estimated to be 0.204 nm and 0.245 nm, respectively. The selected area diffraction patterns in the TiN, interface and HT-9 areas reveal several interesting findings. Consistent with the STEM analysis above, the TiN film grows highly textured with HT-9 substrate in this region evidenced by the highly matched diffraction pattern between the film and the substrate in Fig. 3(e). This suggests that, in this local area, TiN film has established a local epitaxy relationship with HT-9 substrate despite the large lattice mismatch between TiN and HT-9. It is also interesting to note that TiN grows as highly textured polycrystalline with nearly perfect epitaxy with HT-9 substrate. This is consistent with the STEM imaging result in Fig. 3(c). By careful examination of the TiN/HT-9 interfaces in Fig. 3(c), it is clear that the large lattice mismatch has been accommodated by the TiN lattice tilting with the HT-9 lattices. The surprising observation of the nearly perfect epitaxial relationship between HT-9 and TiN is evidenced by the well coupled diffraction patterns from HT-9 and TiN. The epitaxial relationships were determined to be $[111]_{\text{HT-9}} // [110]_{\text{TiN}}$ and $(110)_{\text{HT-9}} // (111)_{\text{TiN}}$. Note that both TiN coating and HT-9 substrate were polycrystalline with large differences in grain size and texture. Thus, it is quite surprising to reveal the epitaxial relationship between TiN and HT-9, which suggests the high-quality growth of the TiN coating on HT-9 and robustness of the coating interface upon irradiation. Note

that the stress assessment at the ceramic-metal interface was only qualitative and the thin TEM foil sample preparation may result in a local stress relaxation compared to the interface stress for the bulk condition.

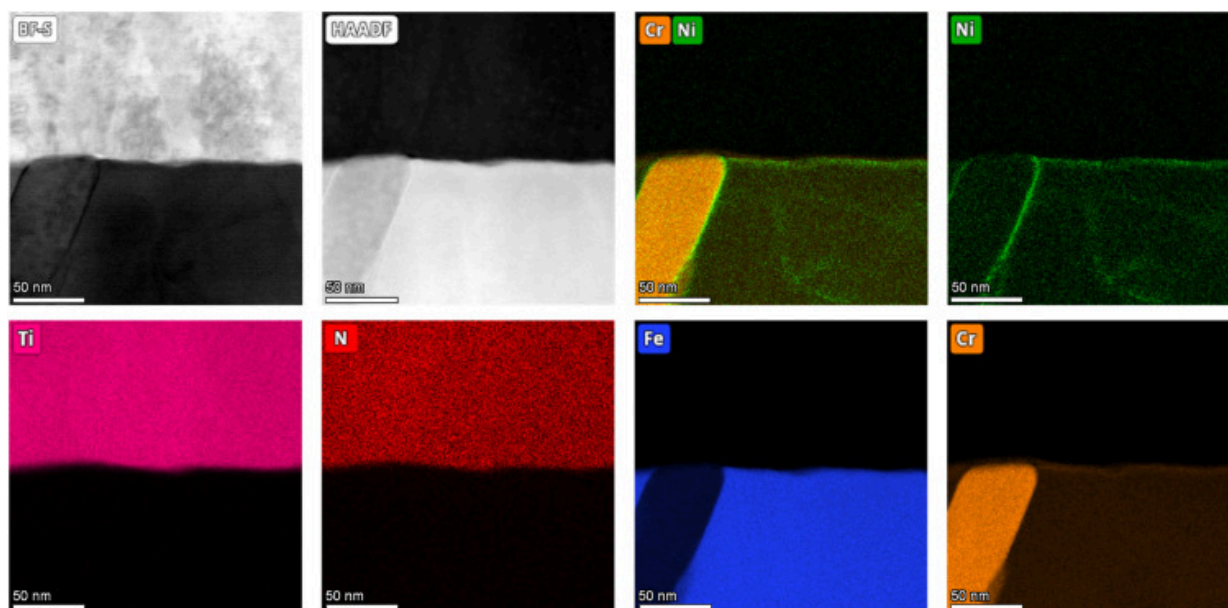


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Fig. 3. For HT-9 alloy coated with 1-mm TiN and irradiated with 5 MeV Fe ions to 50 dpa at 500 °C, STEM bright field image of cross-sectional view of (a) TiN coating on HT-9 substrate, (b) an enlarged view of the TiN/HT-9 interface region (boxed area in a), (c) HR-STEM bright field image showing atomic structure of the ceramic/metal interface from the boxed region in b with HT-9 oriented near [111] zone and TiN coating oriented near [110] zone. The FFT of lattice image in two insets revealed local crystalline orientation. The selected area diffractions from the colored rings in (b) for TiN, interface and HT-9 are shown in (d-f), respectively, with elongated spots indicating higher internal stress and atomic plane distortion.

Energy Dispersive X-ray Spectroscopy (EDS) elemental mapping revealed Ni enrichment at various locations such as HT-9 grain boundaries, the interface between Cr carbide and matrix, and also in the limited region of TiN/HT-9 interface as shown in Fig. 4. Ni segregation was not identified from the unirradiated region of the same HT-9 coupon sample. For ceramic coating bond strength under irradiation, the role of Ni segregation in HT-9 near the ceramic-metal interface needs further investigation. The composition of HT-9 consists of 0.5 wt. % Ni to enhance the alloy's toughness and ductility at low temperature and corrosion resistance when exposed to aqueous or mildly corrosive conditions. It was speculated that the amount of Ni segregation to the ceramic-metal interface under irradiation may be affected by the interface conditions such as the extent of surface Cr oxide formation and surface preparation before coating. Ni enrichment at grain boundaries and carbide surfaces could result in a depletion of Ni in the HT-9 matrix and it may introduce undesired materials property change at the ceramic-metal interface region.



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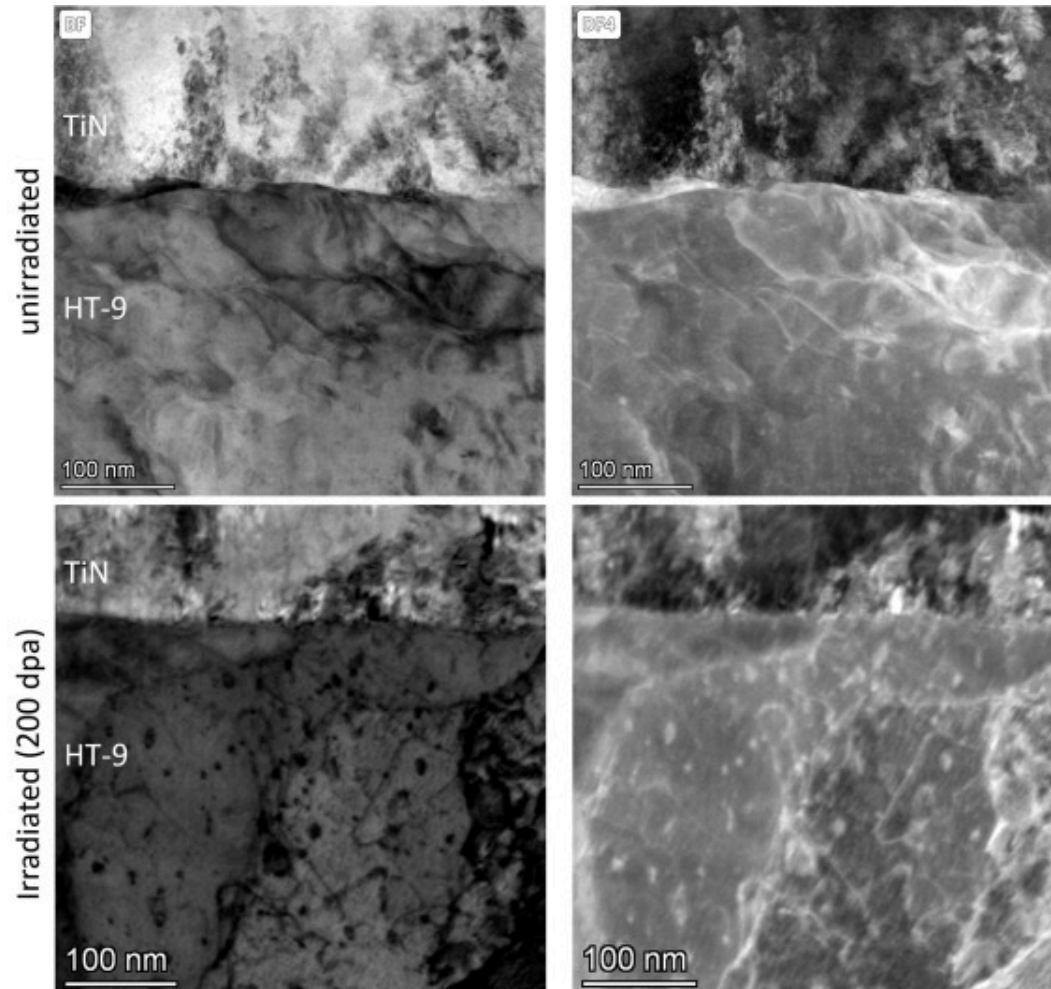
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Fig. 4. TEM bright field (BF) image and high angle annular dark field (HAADF) image of cross-sectional view of TiN and HT-9 alloy interface after irradiation with 5 MeV Fe ions to 50 dpa at 500 °C are shown in upper left with a Cr carbide intersected the HT-9 surface. EDS elemental maps revealed Ni enrichment at grain boundary, carbide-matrix interface and surface in HT-9 as well as a thin Cr oxide surface layer.

3.3. Atomic structure of irradiated TiN-metal interface at 200 dpa

Before discussion of the HR-TEM result at the ceramic-metal interface, it is important to take a look the irradiation induced defects in the interface region at the high dose of 200 dpa. Fig. 5 shows TEM bright field and dark field images of the unirradiated interface and the irradiated interface. Radiation induced dislocation loops are clearly present in the irradiated HT-9 substrate. No radiation induced cavities were identified in TiN and HT-9 and at the interface. For TiN coating with nanosized grains, there is no clear evidence of

radiation induced defects. The nanosized grain texture in TiN coating appears effective in suppressing the radiation induced defects due to extremely high sink density. The lack of void formation in HT-9 at 200 dpa in this work is consistent with literature data [17]. More detailed information on the radiation induced defect development in HT-9 alloy can be found in a review article by Chen [18] and more recent work by Porter et al [19]. There is no significant development of the irradiation induced defects at the TiN and HT-9 interface, and this is important for the TiN coating for nuclear applications.

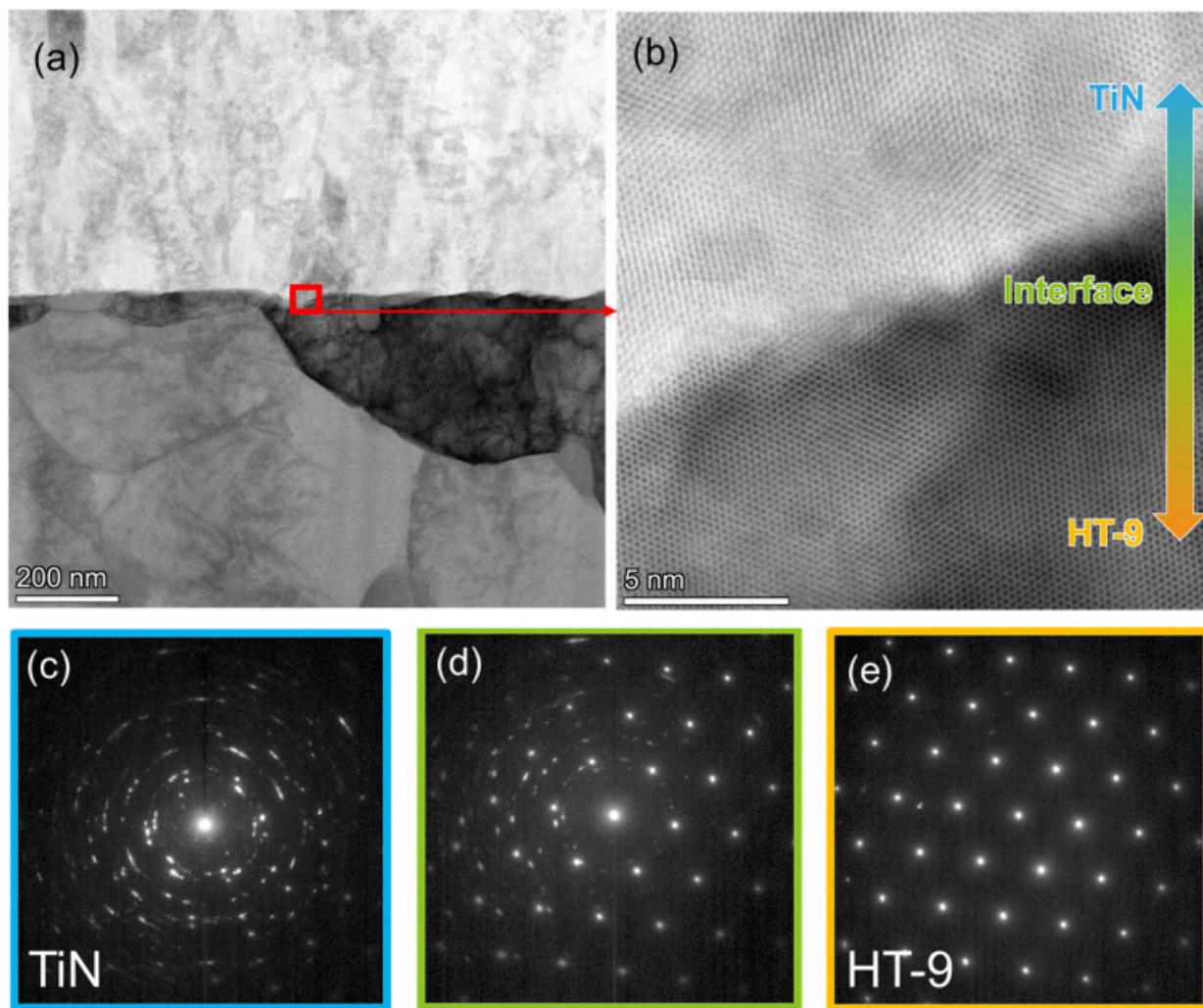


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Fig. 5. TEM bright field (BF) and dark field (DF) images of cross-sectional view of TiN coating and HT-9 alloy interface from the unirradiated region (top) and the irradiated region (bottom, 200 dpa at 500 °C) revealed dislocation loops in the irradiated HT-9 alloy substrate.

Fig. 6(a) with the boxed area, shows where the high-resolution STEM image, shown in Fig. 6(b), was acquired at the ceramic-metal interface. While the irradiation dose was significantly increased from 50 dpa to 200 dpa, the atomic structure at the TiN-Metal interface was not very different and the atomic resolution image of the interface shows that the epitaxial relationships between the TiN and HT-9 could still remain at very high irradiation dose. It appears that the atomic bonding and configuration at the interface between TiN and HT-9, two very different class of materials, is robust and radiation tolerant. No amorphous region was identified in the interface region at 200 dpa. Similar to that of 50 dpa, the selected area diffraction patterns shown in Fig. 6(c), (d) and (e) revealed the highly textured growth of TiN film on HT-9 with epitaxial relationship between the film and substrate as observed above. It is known that during PLD of TiN, the injected Ti and N could have kinetic energy generally in the range of 10–100 eV depending on the laser parameters (e.g., wavelength, pulse duration, and energy), target material properties and local conditions. These kinetic energies of the incoming atoms or ions are sufficient to promote adatom diffusion, strong adhesion and good quality coating on the substrate.



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Fig. 6. TEM bright field (BF) image of cross-sectional view of TiN and HT-9 alloy interface (a) and HR-STEM bright field image showing atomic structure of the ceramic/metal interface from the boxed region in a (b) after irradiation with 5 MeV Fe ions to 200 dpa at 500 °C. The selected area diffractions from the colored regions in (b) for TiN, interface and

HT-9 are shown in (c - e), respectively, with elongated spots indicating higher internal stress and atomic plane distortion.

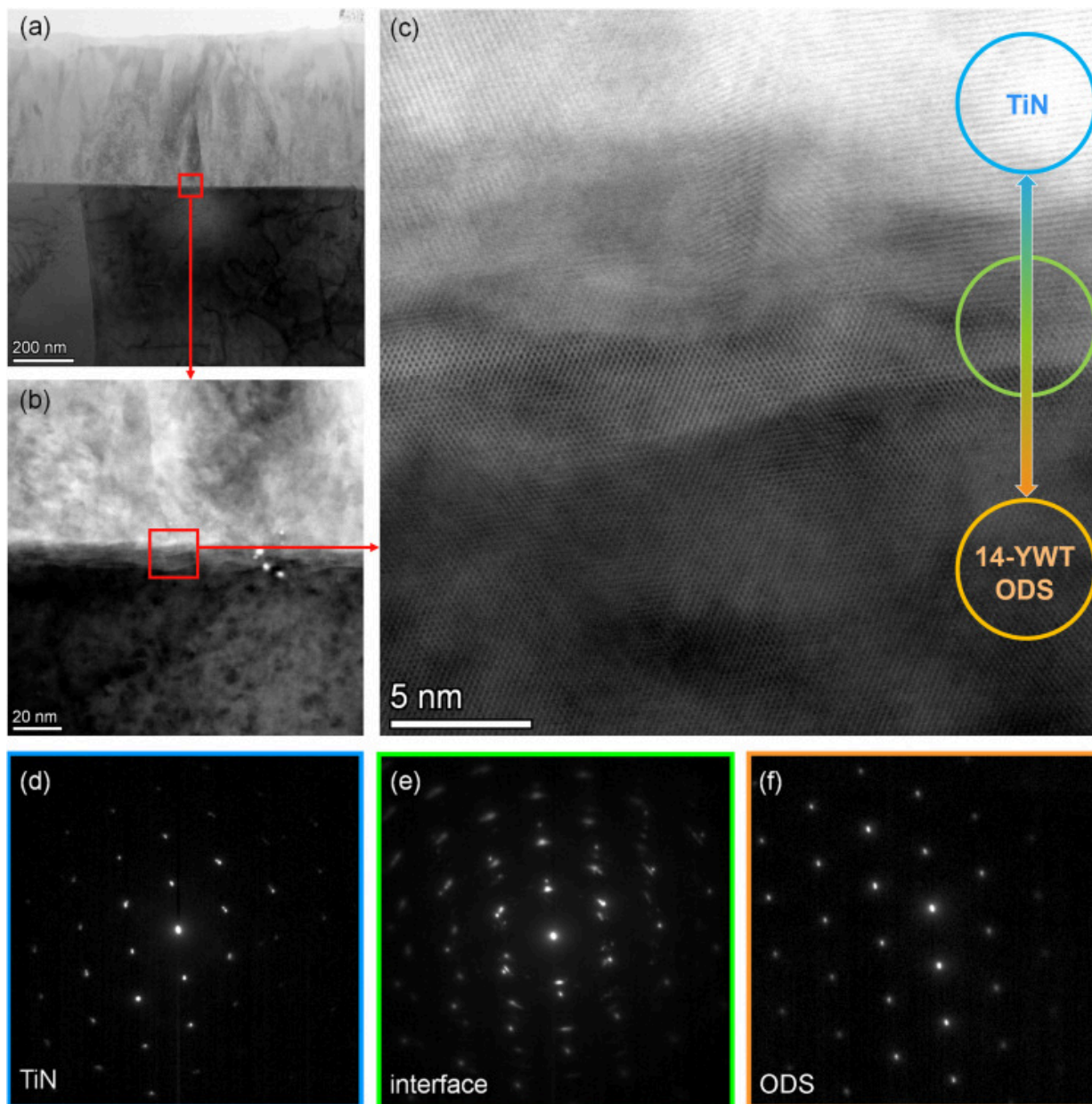
PLD has unique advantages for ceramic coatings and is capable of achieving theoretical densities and target stoichiometry of high-quality ceramic coating at low substrate temperatures even at room temperature. When TiN coating is applied onto the HT-9 surface by PLD, the atomic bonding at the interface typically involves both metallic bonds and covalent bonds. One hypothesis is that the irradiation effect at the ceramic-metal interface may be beneficial to improve the atomic bond strength at the interface. It is generally a concern that the presence of a thin surface oxide on the HT-9 before TiN coating could be detrimental for the ceramic coating. These irradiation effects at the coating-metal interface may disrupt this oxide layer and minimize or eliminate this detrimental effect and promote the atomic bond strength between TiN and HT-9 alloy at the interface. Both theoretical modeling and systematically experimental investigation are required to gain a more comprehensive understanding of the atomic interactions responsible for the interface bond strength during coating deposition and its evolution under irradiation.

3.4. Atomic structure of TiN-metal interface after thermal cycling of 120 - 550 °C

There was a general concern about the interface bonding for ceramic coatings on metal substrates under thermal cycling due to the mismatch in thermal expansion coefficients (TEC). Previous studies on thermal cycling of TiN-coated ODS bar samples showed good performance with no noticeable degradation or interface delamination. However, the HR-TEM investigation of the interface atomic structure after thermal cycling is lacking. It is well-known that TiN and ferritic/martensitic have a significant mismatch in their TEC. During thermal cycling, large amounts of thermal stress build up at the ceramic-metal interface which could potentially promote cracking and debonding. For ceramic coatings on metal substrate it is important to restrict the coating thickness to only a few μm , with small grain sizes of ceramic coating in submicron to offer unique flexibility to

accommodate the large mismatch on TEC. The thin ceramic coating with smaller average grain sizes may behave significantly better than the bulk ceramic counterpart when considering the ceramic-metal interface integrity under thermal cycling.

Fig. 7(a) and (b) are the STEM bright field images revealing the cross-sectional views of interface microstructure for a TiN coated 14-YWT ODS sample. The atomic resolution detail is shown in Fig. 7(c) where ODS alloy substrate is oriented near zone [111], and TiN is oriented off the major zone with different sets of straight-line features corresponding to the projected atomic planes. The colored rings mark the areas where the selected area diffraction was collected and are shown in Fig. 7(d)-(f). It is apparent from both the high magnification image and the atomic-resolution image that the ceramic-metal interface after five thermal cycles in vacuum (10^{-5} – 10^{-6} torr depending on heating temperature), between 120 and 550 °C, appears to be intact without noticeable damage, microcrack or delamination. The orientation relationship between HT-9 and TiN is quite similar to the previous case shown in Fig. 3 for the sample irradiated to 50 dpa, i.e., obvious epitaxial relationship between the TiN coating and the HT-9 substrate with nearly perfect epitaxial growth, evidenced by the distinguished diffraction spots from both TiN and HT-9 regions. Note that the slightly arced diffraction spots were only seen in the selected area diffraction from the interface. This suggests that there are certain degrees of atomic plane bending and twisting in the initial layers of TiN next to the TiN-metal interface resulting from stress introduced by the thermal expansion mismatch between the coating and substrate material. It appears that the thermal stress at the interface was less severe for the thin coating with nanosized grains, such as the 1- μm TiN coating in this work, compared to that of the thick ceramic coating which may lead to debonding or cracking at the ceramic-metal interface when exceeding the limiting conditions. It is speculated that the thin ceramic coating can accommodate large TEC mismatch at ceramic-metal interface by low activation barrier for dislocations and the rotations and sliding of nanocrystalline grains in ceramic at ceramic-metal interface.

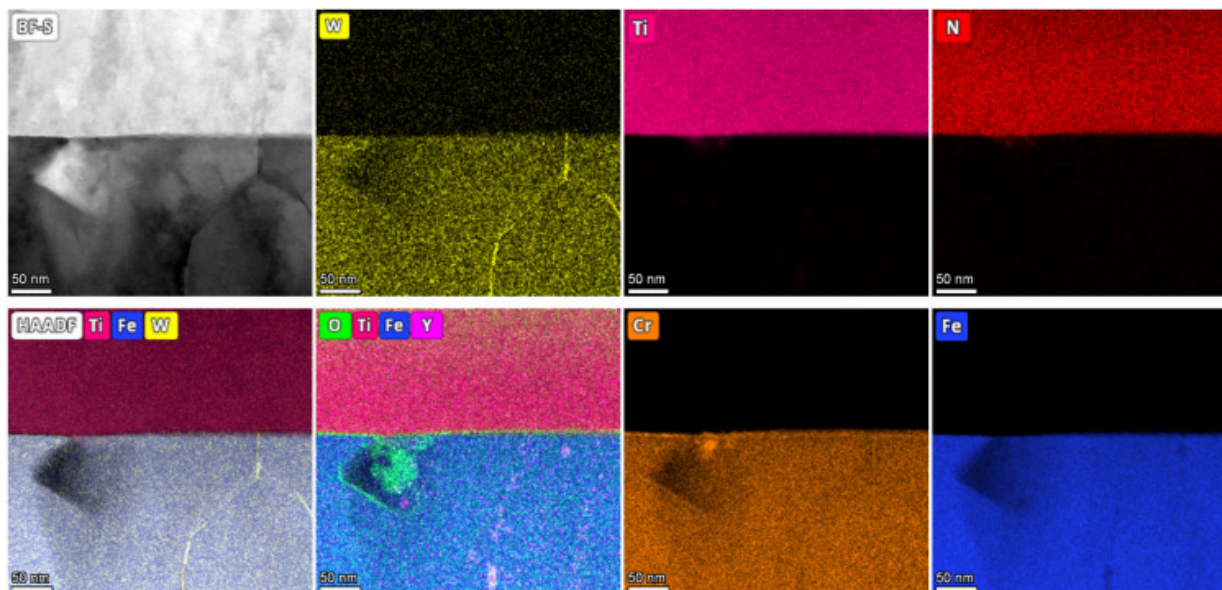


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Fig. 7. For TiN coated 14YWT ODS sample after thermal cycling up to 550 °C with STEM bright field image of cross-sectional view of (a) TiN coating on 14-YWT ODS substrate, (b) an enlarged view of the TiN/ODS interface region (boxed area in a), (c) HR-STEM bright field image showing atomic structure of the ceramic/metal interface from the boxed region in b showing 14-YWT ODS oriented near [111] zone and TiN oriented near [110] zone. The selected area diffractions from the TiN, interface and 14-YWT ODS are shown in (d-f), respectively, with elongated spots indicating higher internal stress and atomic plane distortion.

EDS elemental mapping revealed an unexpected tungsten enrichment to the grain boundaries in the thermally cycled 14-YWT ODS as shown in [Fig. 8](#). The 14-YWT ODS had a nominal W content of 0.5 wt. % in its composition. Thermally induced W segregation is generally considered a detrimental effect since it can lead to the W depletion in the alloy matrix. The effects of this tungsten segregation on the material performance needs further investigation which is beyond the scope of this paper. Small porosities were present in the as-received 14-YWT ODS which was likely introduced from alloy fabrication of this specific alloy batch. More pronounced surface oxide formation in some areas was also noticed in this thermally cycled ODS sample.



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Fig. 8. Bright field (BF) and high angle angular dark field (HAADF) images of cross-sectional view TiN coating and 14-YWT ODS interface after thermal cycling revealed tungsten segregation at grain boundary in 14-YWT and a thin surface Cr oxide layer at the interface.

The key finding from this work is the radiation and thermal stabilities of the TiN-metal interface. The orientation relationship (OR) between TiN and HT-9 appears to play a crucial role in interface stability. The HR-TEM analysis revealed a near-epitaxial relation, specifically $[111]_{\text{HT-9}} // [110]_{\text{TiN}}$ and $(110)_{\text{HT-9}} // (111)_{\text{TiN}}$, which persisted even after irradiation up to 200 dpa. Such a semi-coherent OR reduces the lattice mismatch locally and enables stronger interfacial bonding compared to a random or incoherent interface. A well-defined OR also helps distribute thermal stresses more uniformly across the interface, thereby mitigating the large mismatch in coefficient of thermal expansion between TiN and ferritic steels. During irradiation, a semi-coherent interface with low-

energy misfit dislocations can additionally act as an efficient defect sink, promoting defect recombination and stress relaxation. The persistence of the epitaxial relationship under irradiation and thermal cycling observed in this study suggests that the orientation relationship is a key factor governing the robust bonding and radiation tolerance of the TiN/HT-9 interface. For future work on the TEM analysis of the thermal cycled samples, it will be helpful to compare the grain orientation maps of the TiN side of the interface before and after thermal cycling to investigate the role of grain rotation under thermal expansion mismatch.

The energies of fully coherent interfaces between fcc TiN and bcc Fe were investigated by Yuan et al. using first-principles calculations based on density function theory [20]. The authors found that the Fe(100)/TiN(100) interface with a mismatch of 4.61 % exhibits the lowest interfacial energy of 2.404 J/m². The Fe(110)/TiN(110) interface also exhibits a low mismatch of 4.61 % with a slightly higher interfacial energy of 2.568 J/m². Importantly, they also found that the Fe(110)/TiN(111) interface, which was observed in the HR-TEM analysis in this work, exhibits a large mismatch of 23.1 %, suggesting that a fully coherent interface with this OR would be thermodynamically highly unfavorable compared to a semi-coherent interface. The semi-coherent nature of the Fe(110)/TiN(111) interface may partially explain why the interface of TiN coating on HT-9 was robust against high-dose irradiation.

4. Conclusions

In summary, the detailed HR-TEM study clearly revealed the atomic structural transition at the ceramic-metal interface from the face-centered cubic (FCC) structure of TiN ceramic to the body-centered cubic (BCC) of metal substrate, without discernable microcracks or separation in all four cases (i.e., unirradiated, irradiated to 50 dpa and 200 dpa, and thermal cycled). Radiation effects include dislocation loop formation and Ni segregation to grain boundary and carbide surface with no evidence of cavity formation in both materials and at the interface. These results indicate that the atomic bonding at the TiN and metal interface remained reasonably strong after high doses of irradiation

and thermal cycle tests. The selected area diffraction results suggested obvious highly textured growth of TiN film with epitaxy relationship with the HT-9 substrates for as-deposited sample, irradiated samples and thermal cycled sample. The interfacial lattice tilting of TiN facilitated the epitaxy between the TiN coating and the HT-9 substrate despite the large lattice mismatch. It appears that irradiation may induce stress relaxation at the ceramic-metal interface which could be beneficial to the ceramic-metal interface bond strength. The atomic bond at the interface may consist of a mix of metallic, covalent and ionic bonds. Further experimental and modeling is required to confirm this irradiation-induced beneficial effect on the interface residual stress.

CRediT authorship contribution statement

Yuhan Li: Investigation, Formal analysis, Data curation. **Chao Jiang:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Tiankai Yao:** Resources, Investigation. **Haiyan Wang:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Jian Gan:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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